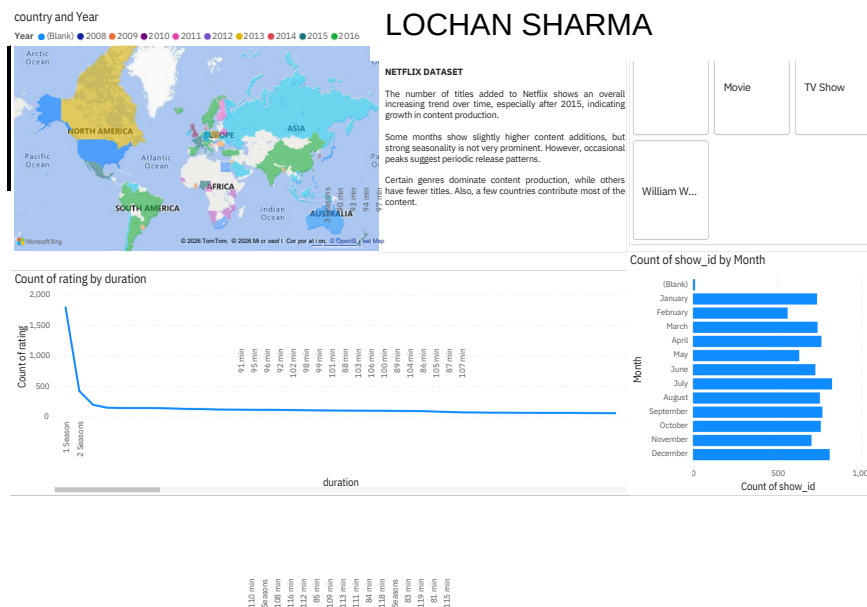




Q1. Identify Trend (1 Mark)

The overall trend is **increasing over time**.

From the dashboard (*page 1*), the number of titles added grows significantly, especially after 2015. This shows a clear upward trend in content production.

Q2. Identify Seasonality (1 Mark)

There is **weak or no strong seasonality**.

Although some months show slightly higher additions, the monthly bar chart does not show a consistent repeating pattern every year. Only minor peaks are visible, indicating **irregular fluctuations rather than strong seasonality**.

Q3. Business Insight (1 Mark)

- The dashboard shows that **some genres and countries contribute less content**, indicating underperformance.
- No direct loss/profit is shown, but **low-performing categories/regions** can be considered weaker areas.

Recommendation:

- Focus on **investing in underperforming genres/regions**
- Increase content production in high-demand areas
- Use audience insights to balance content distribution

Part B

Q4. Why stationarity is important in ARIMA? (1 Mark)

Stationarity is important because ARIMA assumes that:

- Mean, variance, and patterns remain constant over time

If data is **non-stationary**, the model cannot correctly identify patterns, leading to inaccurate forecasts.

So, we apply **differencing** to make data stationary before using ARIMA.

Q5. ARIMA vs SARIMA (1 Mark)

Feature	ARIMA	SARIMA
Full Form	AutoRegressive Integrated Moving Average	Seasonal ARIMA
Handles Trend	Yes	Yes
Handles Seasonality	No	Yes
Components	(p, d, q)	(p, d, q)(P, D, Q, s)

When to use SARIMA?

Use SARIMA when data shows **seasonal patterns (e.g., yearly, monthly cycles)**.

Q6. Case Study Answer (1 Mark)

Model Choice:

➡ SARIMA

Justification:

- Data shows **increasing trend**
- There is **clear yearly seasonality** (Nov–Dec spikes every year)
- Pattern repeats consistently

ARIMA cannot handle seasonality, but SARIMA can.

What 'S' represents in SARIMA:

- **S = Seasonal component**
- It captures repeating patterns over a fixed period (e.g., 12 months in a year)